Experiment 6

Aim

Create a stored procedure transfer_employee(emp_id, new_dept_id) with validation and error handling. Implement triggers for salary validation and audit logging. Test edge cases.

Source Code

1. Audit Table

```
CREATE TABLE Emp_Audit (
    audit_id     INT AUTO_INCREMENT PRIMARY KEY,
    emp_id       INT,
    action       VARCHAR(10),
    old_salary   DECIMAL(10,2),
    new_salary   DECIMAL(10,2),
    old_dept     INT,
    new_dept     INT,
    action_time  TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    performed_by VARCHAR(50)
);
```

2. Triggers

```
-- Salary validation trigger
DELIMITER $$
CREATE TRIGGER trg_salary_validate
BEFORE INSERT ON Employee
FOR EACH ROW
BEGIN
    IF NEW.salary <= 0 THEN
```

```
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Salary must be greater than zero.';
    END IF;
END$$
DELIMITER ;
-- Audit logging trigger for UPDATE
DELIMITER $$
CREATE TRIGGER trg_emp_audit_update
AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
  IF OLD.salary <> NEW.salary OR OLD.dept_id <> NEW.dept_id THEN
    INSERT INTO Emp_Audit (emp_id, action, old_salary, new_salary,
                           old_dept, new_dept, performed_by)
    VALUES (OLD.emp_id, 'UPDATE', OLD.salary, NEW.salary,
            OLD.dept_id, NEW.dept_id, USER());
  END IF;
END$$
DELIMITER ;
```

3. Stored Procedure

```
sql
DELIMITER $$
CREATE PROCEDURE transfer_employee(
  IN p_emp_id      INT,
  IN p_new_dept_id INT
)
BEGIN
  DECLARE v_old_dept   INT;
  DECLARE v_emp_exists INT DEFAULT 0;
  DECLARE v_dept_exists INT DEFAULT 0;
```

```
-- Error handler
DECLARE EXIT HANDLER FOR SQLEXCEPTION
BEGIN
    ROLLBACK;
    SELECT 'Error: Transaction rolled back due to an exception.' AS message;
END;

START TRANSACTION;
-- Validation: employee exists
SELECT COUNT(*) INTO v_emp_exists FROM Employee WHERE emp_id = p_emp_id;
IF v_emp_exists = 0 THEN
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT = 'Employee does not exist.';
END IF;

    -- Validation: new department exists
    SELECT COUNT(*) INTO v_dept_exists FROM Department WHERE dept_id =
p_new_dept_id;
    IF v_dept_exists = 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Department does not exist.';
    END IF;

    -- Get current department
    SELECT dept_id INTO v_old_dept FROM Employee WHERE emp_id = p_emp_id;

    -- Validation: same department
    IF v_old_dept = p_new_dept_id THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Employee is already in the specified department.';
```

Output:

```
+----------------------------------------+
| message                                |
+----------------------------------------+
| Employee 1 transferred from dept 1 to 3 |
+----------------------------------------+
+------------------------------------------------------+
| message                                              |
+------------------------------------------------------+
| Error: Transaction rolled back due to an exception. |
+------------------------------------------------------+
+------------------------------------------------------+
| message                                              |
+------------------------------------------------------+
| Error: Transaction rolled back due to an exception. |
+------------------------------------------------------+
+------------------------------------------------------+
| message                                              |
+------------------------------------------------------+
| Error: Transaction rolled back due to an exception. |
+------------------------------------------------------+
ERROR 1644 (45000) at line 182: Salary must be greater than zero.
```

